

Operational Excellence Standard Operating Procedure

Document ID	OE-SOP-001
Version	3.2
Owner	Group Operational Excellence
Classification	Internal

1. Purpose

This Standard Operating Procedure defines the minimum operating discipline expected of every PETRONAS asset, business unit and operating site within the Group. It establishes a common language for operational performance and provides the basis for cross-business benchmarking.

All sites are expected to comply with this SOP regardless of business unit, geography, or asset type.

2. Scope

This SOP applies to all PETRONAS-operated facilities including Upstream offshore platforms and onshore fields, Downstream refineries and petrochemical plants, Gas & New Energy LNG plants, Trading desks, Retail clusters, and Logistics terminals.

It does not apply to non-operated joint ventures or facilities operated under third-party operating models.

3. Operating Discipline Pillars

3.1 Throughput Reliability. Every operated site shall report a daily throughput figure against an approved throughput target. Sites operating below 90% of approved target for three consecutive days shall trigger an Operational Performance Review.

3.2 Energy Stewardship. Energy consumption per unit of throughput shall be tracked daily and reported monthly to Group Operational Excellence. Sites are expected to deliver year-on-year energy intensity improvements aligned with the Group's Net Zero 2050 pathway.

3.3 Emissions Accounting. Scope 1 emissions in tCO₂e shall be calculated daily using approved emission factors for the asset type. Reporting cadence is monthly with annual external assurance.

3.4 Asset Integrity. Every critical asset shall have an inspection cadence registered in the Asset Register. Overdue inspections shall be escalated to the Asset Integrity Manager within 5 business days.

4. Reporting Standards

All operational data shall flow into the Group Operations Data Platform on Databricks Unity Catalog. Each business unit is responsible for daily ingestion of throughput, uptime, energy and emissions data into the operations_daily Delta table.

Data quality expectations: completeness >= 98% per site per month, latency <= 24 hours from event time to availability in the platform, and column-level lineage maintained through Unity Catalog.

5. Roles & Responsibilities

Site Operations Lead: Owns daily data submission and first-line investigation of variances against target.

Business Unit Operations Manager: Reviews weekly performance, approves variance explanations, sponsors corrective actions.

Group Operational Excellence: Maintains the SOP, the Group dashboard, and benchmarks performance across business units.

6. Performance Review Cadence

Daily: Site Operations huddle (15 min) to review overnight throughput, uptime and exceptions.

Weekly: Business Unit Operations Review covering trailing 7-day performance and forward-week look-ahead.

Monthly: Group Operations Council reviewing Group-wide KPIs, energy intensity trend, and emissions performance.

Quarterly: External assurance review for emissions data ahead of regulatory submissions.

7. Definitions

Throughput Target: The approved throughput plan for the operating period as agreed by the BU Operations Manager.

Uptime: Hours an asset is producing within design tolerances divided by total hours in the period.

Energy Intensity: MWh of energy consumed per unit of throughput, normalised to a reference site within the same site type.

Tier-1 Variance: A throughput shortfall exceeding 15% versus target sustained for more than 24 hours.

This document is synthetic and prepared solely for the PETRONAS AI Enterprise Adoption Workshop. It does not represent actual operational policies of PETRONAS or its subsidiaries.